

Constant acceleration

1. An Eddie Stobart lorry starts from rest and accelerates uniformly, at 1.5 ms^{-2} , until it attains a speed of 30 ms^{-1} . Find:
 - (a) the distance the lorry travels during this motion
 - (b) the time taken.
2. A pigeon flies in a straight line uniformly increasing its velocity from 2 ms^{-1} to 18 ms^{-1} in a time interval of 10 s. Find:
 - (a) the acceleration of the pigeon during this time
 - (b) the distance travelled.
3. A cheetah can accelerate from rest to 30 ms^{-1} in a distance of 25 m. Find the acceleration if we assume it to be constant.
4. The manufacturer of a new electric car claims that it can accelerate from rest to 90 km/h in 10 seconds. Find the acceleration (assumed constant).
5. In travelling the 70 cm along a rifle barrel, a bullet uniformly accelerates from its initial state of rest to a muzzle velocity of 210 ms^{-1} . Find:
 - (a) the acceleration involved
 - (b) the time for which the bullet is in the barrel.
6. According to the Highway Code, a car travelling at 20 ms^{-1} requires a minimum braking distance of 30 m. What retardation is this and what length of time will it take?
7. A motorbike is initially at rest at a set of traffic lights. The motorbike moves away from the traffic lights in a straight line, accelerating at 4 ms^{-2} . Find how far the motorbike travels during the third second of motion.
8. At time $t=0$, a particle is projected from a point P with an initial velocity of 10 ms^{-1} . The particle moves along a straight line with a constant acceleration of -2 ms^{-2} .
 - (a) Find the displacement of the particle from P when $t=7$ seconds.
 - (b) How far from P does the particle come to instantaneous rest and what is the value of t then?
 - (c) Find the distance travelled by the body during the time interval $t=0$ to $t=7$ seconds.
9. A car is being driven along a road at a steady 90 km/h when the driver suddenly notices that there is a fallen tree blocking the road 65 metres ahead. The driver immediately applies the brakes giving the car a constant retardation of 5 ms^{-2} .
 - (a) How far in front of the tree does the car come to rest?
 - (b) If the driver had not reacted immediately and the brakes were applied one second later, with what speed (in km/h) would the car have hit the tree?
10. An underground train travels along a straight piece of track between Golders Green station and Hampstead station. The train starts from rest at Golders Green and accelerates at 1.25 ms^{-2} until it reaches a speed of 20 ms^{-1} . It then travels at this steady speed for a distance of 1.56 km and then decelerates at 2 ms^{-2} to come to rest at Hampstead station. Find:
 - (a) the distance from Golders Green station to Hampstead station
 - (b) the total time taken for the journey
 - (c) the average speed for the journey.
11. A shark is swimming at constant velocity when it notices a surfer ahead struggling to get back on his board. The shark accelerates uniformly towards the surfer and after 2 seconds passes a stationary piece of debris (10 m from where the shark first noticed the surfer). The shark's speed at this point is 7 ms^{-1} . If the shark reaches the surfer 3 seconds later, find (assuming the surfer did not move):
 - (a) the acceleration of the shark
 - (b) the shark's speed when it first noticed the surfer.
12. A, B and C are three points which lie in that order on a straight road with $AB = 95 \text{ m}$ and $BC = 80 \text{ m}$. A van is travelling along the road in the direction ABC with constant acceleration $a \text{ ms}^{-2}$. The van passes through A with speed $u \text{ ms}^{-1}$, reaches B five seconds later, and C two seconds after that. Find the values of u and a .

13. A reckless driver is travelling at a constant 90 km/h along a road with a 60 km/h speed limit and doesn't notice passing a stationary police car. Two seconds later the police car sets off in pursuit, accelerating at a uniform 6 ms^{-2} . How far does the police car need to travel before catching up with the reckless (and unobservant) driver?

Objects falling under gravity

14. A ball is kicked vertically upwards from ground level with an initial speed of 14 ms^{-1} . Find the height above ground level of the highest point reached and the total time for which the ball is in the air.
15. A ball is held 1.6m above a concrete floor and released. The ball hits the floor and rebounds with half the speed it had just prior to impact. Find the greatest height the ball reaches after:
- (a) the first bounce
 - (b) the second bounce.
16. A body is projected vertically upwards from ground level at a speed of 49 ms^{-1} . Find the length of time for which the body is at least 78.4m above the ground.
17. A particle is projected vertically upwards with a speed of 14 ms^{-1} . Find the height of the particle above the level of projection after:
- (a) 1 second of motion
 - (b) 2 seconds of motion
- Find the distance travelled by the body in the 2nd second of motion.
18. Two stones are thrown from the same point on a cliff edge at the same time, one vertically upwards with speed 30 ms^{-1} , and the other vertically downwards at 30 ms^{-1} . Find how far apart the stones are after 3 seconds.